3 Determination of the enthalpy change when citric acid reacts with sodium hydrogencarbonate

Citric acid, $C_6H_8O_7$, is found in citrus fruit such as lemons and limes. It is a triprotic (tribasic) acid – one mole of citric acid reacts with three moles of sodium hydrogencarbonate.

equation 4 $C_6H_8O_7(aq) + 3NaHCO_3(s) \rightarrow C_6H_5O_7Na_3(aq) + 3CO_2(g) + 3H_2O(l)$ ΔH_1

FA 4 is 0.080 mol dm^{-3} citric acid, $C_6H_8O_7$.
FA 5 is solid sodium hydrogencarbonate, $NaHCO_3$.

(a) Calculate the minimum mass of sodium hydrogencarbonate that will react completely with the citric acid in 50.0 cm^3 of **FA 4**.
[A_r: Na, 23.0; H, 1.0; C, 12.0; O, 16.0]

minimum mass of $NaHCO_3$ = ... [1]

(b) Determination of the molar enthalpy change of reaction, ΔH_1

In this experiment, you will measure the temperature of the contents of a polystyrene cup at timed intervals, both before and after an excess of **FA 5** is added. You will analyse your results graphically to obtain an accurate value for the temperature change caused by the reaction.

You will use this value to calculate the heat change, q, for the experiment and hence determine a value for the molar enthalpy change of the reaction, ΔH_1.

In the space provided on page 7, prepare a table in which to record for your experiment:

- all values of temperature, T, to an appropriate level of precision
- all values of time, t, recorded to the nearest 0.5 min.

It is important that you measure each temperature at the specified time.

Procedure

1. Measure 50.0 cm^3 of **FA 4** using a 50.0 cm^3 measuring cylinder.

2. Place one polystyrene cup inside a glass beaker.

3. Transfer 50.00 cm^3 of **FA 4** into the polystyrene cup.

4. Carefully stir the **FA 4** in the polystyrene cup with the thermometer. Read and record the temperature, *T*. Start the stopwatch (t = 0.0 min). The stopwatch must be left to run for the rest of the experiment.

5. Continue to stir **FA 4**. Read and record *T* every 0.5 minute for 1.5 minutes.

6. At exactly 2.0 minutes, transfer all the solid **FA 5** (an excess) to the polystyrene cup. Stir the mixture thoroughly but do not read *T*.

7. Continue to stir the mixture. Read and record *T* at t = 2.5 min and every 0.5 min until the temperature reaches a minimum, and then every minute for a further 4 minutes.

(i) Results

[3]

(ii) Plot a graph of the temperature on the y-axis, against time, *t,* on the x-axis on the grid in Fig. 3.1.

Draw a best-fit straight line taking into account all of the points before t = 2.0 min.

Draw another best-fit straight line taking into account all of the points after the temperature of the mixture has started to rise steadily.

Extrapolate (extend) both lines to t = 2.0 min. [3]

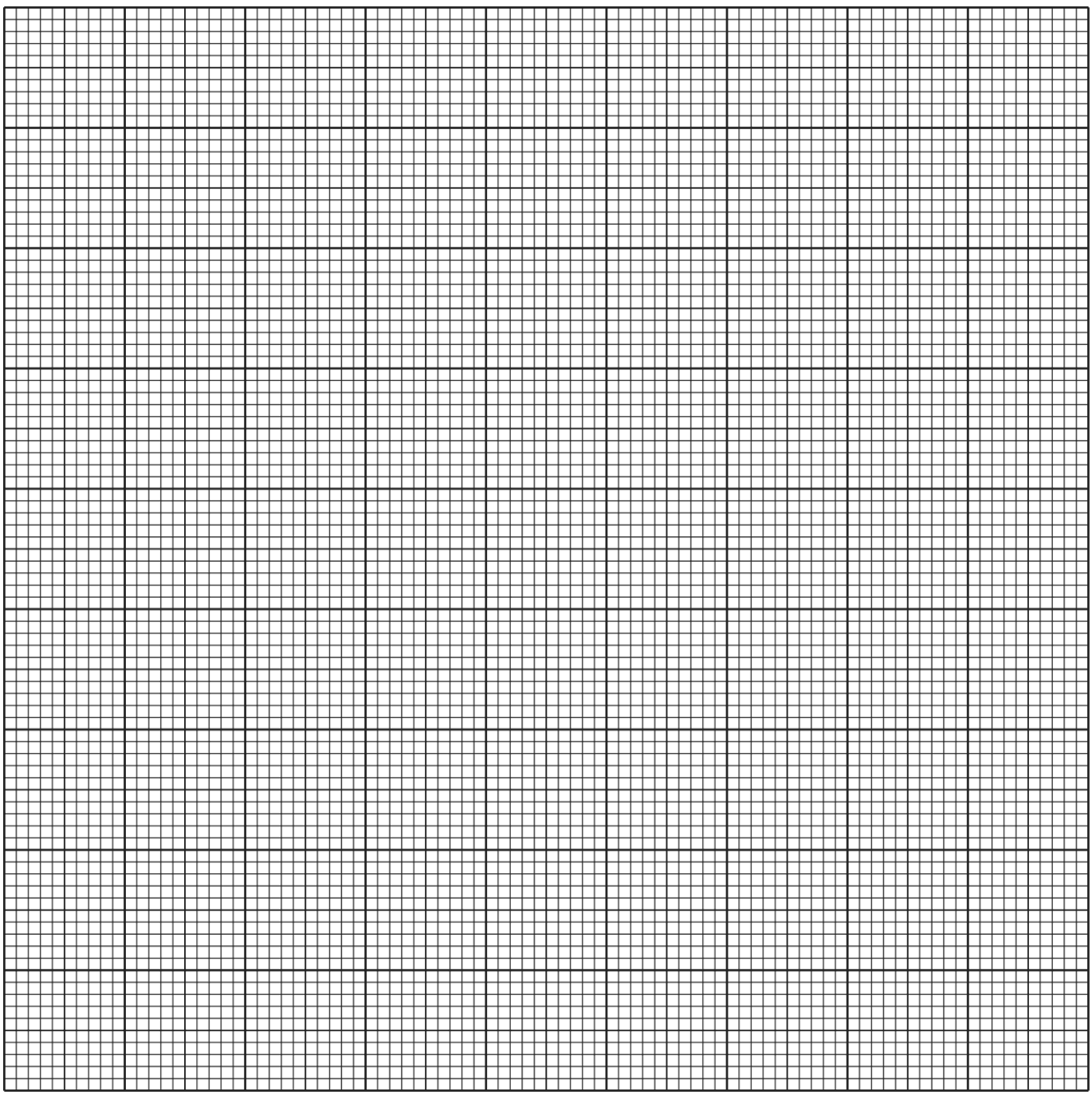

Fig. 3.1

(iii) From your graph, read the minimum temperature, T_{min}, and the maximum temperature, T_{max}, at t = 2.0 min. Record these values in the spaces provided.

Deduce the temperature change, ΔT, at t = 2.0 min.

T_{min} = ...

T_{max} = ...

ΔT = ...

[1]

(iv) Calculate the heat change, q, for your experiment using the value you calculated in **(b)(iii)**.

Assume that the specific heat capacity of the reaction mixture is 4.18 J g^{-1} K^{-1}, and that the density of the solution is 1.00 g cm^{-3}.

q = ... [1]

(v) Determine the molar enthalpy change, ΔH_1, for the reaction in equation 4. The sodium hydrogencarbonate is in excess.

Include the sign of ΔH_1 in your answer.

ΔH_1 = ... [5]

(c) **(i)** The maximum temperature change, ΔT_{max}, can also be determined by direct measurement of the initial temperature and the lowest temperature reached only.

Explain why the graphical method is likely to give a more accurate value for ΔT_{max} than the direct measurement method.

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..... [1]

(ii) What would be the expected change in temperature obtained in **(b)(iii)** if the volume of citric acid used was halved? Explain your answer.

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..... [2]

[Total: 17]